

Numerical Expression (Ciselny Vyras)

Exam Task — Java OOP

Overview

The goal is to create a program that can evaluate arbitrarily complex numerical expressions — sequences of numbers and mathematical operations. The program has simplified rules:

- Input contains only decimal numbers (0-9 digits, decimal separator)
- Operations: addition (+), subtraction (-), multiplication (*), division (/)
- Numbers and operations always alternate; separated by spaces
- No parentheses
- Expression must start and end with a number
- Invalid expressions throw an exception
- Division by zero does not need to be handled

The program evaluates in two modes:

- **With priority** — multiplication and division have precedence over addition and subtraction
- **Without priority** — operations are evaluated left to right

Transformation Process

The expression is read as a String from console. It must be transformed into a collection of parts (CastVyrasu) — alternating Cisko and Operacia objects. The first element is always at index 0 (a number), followed by an operation at index 1, and so on.

During transformation, validate:

- Expression starts and ends with a number
- No two operations or two numbers in a row
- Each number is a valid decimal number
- Each operation symbol is valid (+, -, *, /)

If invalid, throw an exception.

Calculation Step

Each step finds the first suitable operation (by strategy), computes it, and replaces the two numbers and the operation in the collection with the result (a single Cisko). Steps repeat while the collection has more than one element.

Input / Output Format

On startup, program prints:

```
Zadaj ciselny vyraz >
```

After reading input, prints the expression in normalized form (3 decimal places, spaces around operators):

```
Zadany ciselny vyraz: 5,000 + 1,430 * 5,450
```

After each step:

```
Krok 1: 5,000 + 7,794  
Krok 2: 12,794
```

Both modes are shown (with priority first, then without). Separator between modes:

```
- - - - -
```

If expression is invalid:

```
Ciselny vyraz nie je zapisany spravne.
```

Class Descriptions

CastVyrazu (2 pts) — interface for all expression parts

- getReprezentacia(): String — returns text representation

Cislo (4 pts) — immutable class representing a number

- new(hodnota: double)
- getReprezentacia(): String — returns value rounded to 3 decimal places
- getHodnota(): double

Operacia (7 pts) — immutable class representing a math operation

- new(operacia: char)
- vypocitaj(laveCislo: double, praveCislo: double): double — computes result; throws exception if invalid operation
- getReprezentacia(): String — returns the operation symbol as String
- jeHladanaOperacia(hladaneOperacie: char[]): boolean — returns true if this operation is in the array

VyberacOperacie (2 pts) — interface for operation selection strategy

- dajIndexOperacie(vypocet: Vypocet): int — returns index of operation to compute next

VyberacBezPriority (3 pts) — returns first available operation (left to right)

- dajIndexOperacie (2 pts) — returns index of first operation; throws exception if none found

VyberacSPrioritou (3 pts) — returns first * or / operation; if none, falls back to first + or -

- dajIndexOperacie (2 pts) — throws exception if no operation found

Vypocet (14 pts) — manages the calculation

- new(vyraz: Iterable<CastVyrazu>, vyberacOperacie: VyberacOperacie) (2 pts)
- vykonajKrok() (5 pts) — executes one calculation step per algorithm. Returns true if step was performed (operation computed); false if expression is already fully computed
- getVyrazString() (2 pts) — returns current expression as formatted string
- najdiPrvuOperaciju(hladaneOperacie: char[]) (4 pts) — finds operation matching filter; returns its index in Optional, or empty Optional if not found

CiselnyVyraz (12 pts) — immutable class, represents full expression

- vytvorZRetazca(vyrazyVTexte: String): CiselnyVyraz (7 pts) — static method; transforms string into collection of parts, validates, creates instance
- vyries(vyberacOperacie: VyberacOperacie) (3 pts) — solves expression step by step, prints output per format

Main (3 pts) — contains main method:

- Loops: reads input, if 'koniec' exits, otherwise solves with both strategies
- If exception thrown, prints error message; program continues

Points Summary

Class	Points
CastVyrazu (interface)	2
Cislo	4
Operacia	7
VyberacOperacie (interface)	2
VyberacBezPriority	3

VyberacSPrioritou	3
Vypocet	14
CiselnyVyras	12
Main	3
TOTAL	50